Training language models to follow instructions with human feedback

Presenters

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Authors: Ouyang, L. et al. (2022)
Venue: NeurIPS 35
Link: `https://arxiv.org/abs/2203.02155`

Key Takeaway

This paper proposes a Reinforcement Learning from Human Feedback (RLHF) based fine-tuning method to make a large language model (LLM) more helpful, honest, and harmless.

Motivation & Problem

Problem

- Alignment: a model is helpful, honest, and harmless.
- LLMs are trained to predict the next token, while users hope them to follow the instructions safely and helpfully.

Gap in prior work

- **Limit of Scaling**: Making models bigger does not inherently improve instruction following.
- **Limit of NLP datasets**: Public NLP datasets do not reflect real-world LLM usage patterns.

Core Idea (High-Level)

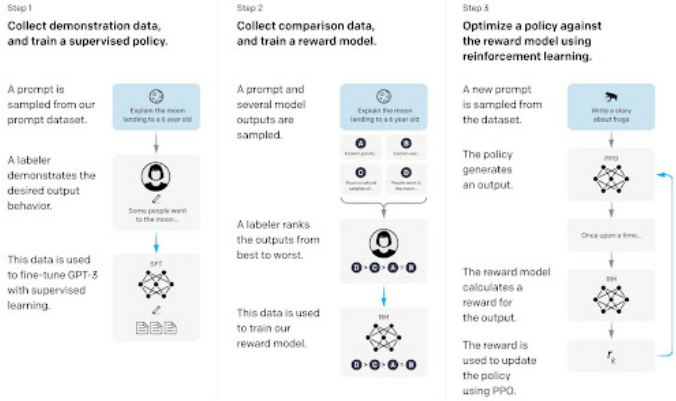
Main contribution

- Use RLHF to fine-tune LLMs — producing more aligned models that are also more lightweight.
- Human preferences serve as a reward signal to guide fine-tuning.

Three-step mental model

- **Supervised Fine-Tuning (SFT)**: Teach LLM to respond via human-written demonstrations.
- **Reward Modeling (RM)**: Train a model to predict which LLM response humans prefer.
- **Reinforcement Learning (RL)**: Use PPO to optimize the LLM against the reward model.

Method — The 3-Step RLHF Pipeline



Step 1: SFT

Fine-tune GPT-3 on ∼13k human demonstrations (labeler + customer prompts)

Step 2: Reward Model

Initialize from SFT; Train on human preference rankings; outputs scalar reward

Step 3: PPO

Optimize policy to maximize RM reward via RL

Key Assumption

Human preference rankings are a reliable and generalizable alignment signal

Method — Step 1: Supervised Fine-Tuning (SFT)

Why

GPT-3's next-token objective does not teach instruction-following. SFT teaches the model to imitate human responses in various tasks.

How

- Dataset: $\sim$13k prompts (written labels + real API submissions)
- Training: 16 epochs, cosine LR decay, residual dropout 0.2
- Model selected by **RM score** on the validation set (not raw loss)

Output

A capable instruction-following **SFT baseline** — the starting point for RLHF.

Method — Step 2: Reward Model (RM) Training

Why

Ranking output is faster and more scalable than writing demonstrations. Train a 6B model to predict which output a human prefers (scalar reward).

How

- Labelers rank $K = 4$–9 outputs per prompt ($\sim$33k prompts)
- **Key innovation:** batch all $\binom{K}{2}$ pairs in one batch element — avoids overfitting, boosts efficiency
- Normalize: set mean reward of human demos to 0 before RL

Pairwise Ranking Loss

$$\mathcal{L}(\theta) = -\frac{1}{\binom{K}{2}} \mathbb{E}_{(x, y_w, y_l)}[\log \sigma(r_\theta(x, y_w) - r_\theta(x, y_l))]$$

where y_w is preferred over y_l for prompt x

Method — Step 3: Reinforcement Learning via PPO

Standard PPO

- Maximize RM reward while staying close to SFT policy
- Per-token **KL penalty** ($\beta = 0.02$) prevents reward hacking:

$$r = r_\theta(x, y) - \beta \log \frac{\pi^{\mathbf{RL}}}{\pi^{\mathbf{SFT}}}$$

- Value function from 6B RM; 256k episodes

PPO-ptx — Final InstructGPT

- Mixes pretraining gradients to fix NLP regressions:

$$\mathcal{J} = \underbrace{r - \beta\,\mathrm{KL}}_{\text{PPO}} + \underbrace{\gamma \log \pi^{\mathrm{RL}}(x)}_{\text{pretrain loss}}$$

- $\gamma = 27.8$; pretrain data ratio $= 8 : 1$
- **This is the final InstructGPT model**

Models Evaluated

Three scales: **1.3B**, **6B**, and **175B** parameters — all using GPT-3 architecture

Results — Evaluation Setup

Human Evaluation (Primary)

- Held-out prompts from unseen users + real GPT-3 API prompts (excluded from training)
- Head-to-head vs. 175B SFT baseline using **win rate**
- 1–7 Likert quality ratings + metadata (instruction following, constraint adherence, hallucinations)

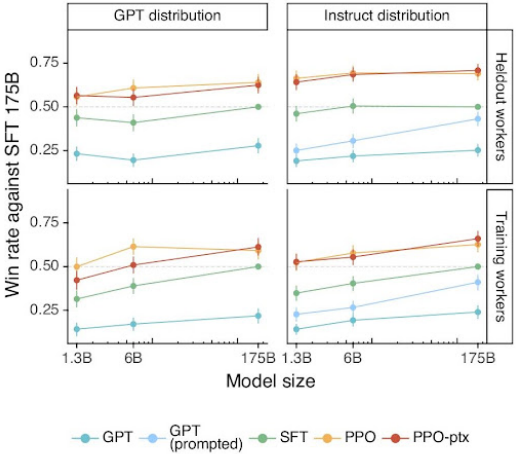
Truthfulness & Safety

- **TruthfulQA** — truthful and informative responses
- **RealToxicityPrompts** — Perspective API toxicity + human eval
- **Winogender / CrowS-Pairs** — gender and social bias

Capability Benchmarks (Alignment Tax Check)

To measure alignment tax, we evaluate on standard NLP datasets: Commonsense reasoning (HellaSwag), Reading comprehension (DROP, SQuAD etc), and translation (WMT 2015 Fr→En)

Results — Human Preference Win Rates



85% win rate
InstructGPT 175B vs. GPT-3 175B

71% win rate
InstructGPT vs. few-shot GPT-3

1.3B beats 175B
InstructGPT beats GPT-3
(100× smaller!)

Win rates vs. 175B SFT baseline

Results generalize to held-out labelers

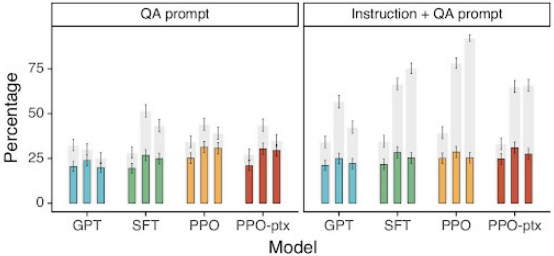
Results — Truthfulness & Safety

Truthfulness (TruthfulQA)

- PPO shows **significant improvements** in truthfulness & informativeness vs. GPT-3
- Improvement holds by default (no special instruction needed)
- Reduced hallucinations on API tasks: **21% vs. 41%** (∼50% drop)

Toxicity & Bias

- 25% fewer toxic outputs (when prompted respectfully)
- Without respectful prompt: similar to GPT-3
- Bias (Winogender/CrowS-Pairs): **no significant improvement**



TruthfulQA: gray = truthful; colored = truthful & informative

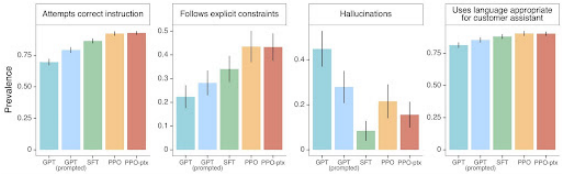
Results — Helpfulness & Alignment Tax

Metadata Improvements vs. GPT-3

- Follows explicit constraints: **45% vs. 20%**
- Attempts correct instruction: **85% vs. 60%**
- More appropriate as a customer assistant

Alignment Tax & Fix

- Pure PPO regresses on NLP tasks (SQuAD v2: 64.3 → 43.7 F1)
- **PPO-ptx recovers:** SQuAD v2 → 59.9; HellaSwag **exceeds** GPT-3
- Generalizes to non-English & code (<4% training data)



InstructGPT follows instructions better and hallucinates less

Limitations

Labeler Representativeness

- Only $\sim$40 contractors — mostly English-speaking, US or SE Asia
- Inter-annotator agreement $\approx$ 73%; disagreement on many examples
- Model aligns to *this group's* preferences, not universal human values

Residual Model Failures

- Still hallucinates and over-hedges on simple questions; confused by false-premise prompts
- Can generate **toxic** content without prompting and follows harmful instructions if explicitly asked

Incomplete Alignment Tax Fix

- PPO-ptx reduces but does **not fully eliminate** regressions on SQuaD, DROP and translation
- Pretraining mix can reintroduce undesirable behaviors from pretraining data

Discussion Questions

1. The reward model reflects preferences of a **small, non-representative labeler pool**. How fragile is this for diverse global users or safety-critical domains like smart-grid operators?
2. RLHF makes models more instruction-following — but also **more willing to follow harmful instructions**. How should this trade-off be managed in energy infrastructure where mis-commands could cause outages?
3. Could RLHF align LLMs for **smart-grid tasks** (demand response, fault diagnosis, load forecasting)? What challenges arise collecting expert preference data from engineers vs. general labelers?